

SAFETY DATA SHEET

BRSMMH — Positive Electric Adhesive Dry Powder

Prepared in accordance with the Globally Harmonized System of Classification and Labelling of Chemicals (GHS)

Section 1: Identification

Product identifier: BRSMMH

Product type: Positive Electric Adhesive Dry Powder (positively charged mixed-layered metal hydroxide crystal colloid)

Executive standard: Q/PSL038-2003

Recommended use: Drilling fluid stabilizer and wellbore stabilizer, especially for complex formations and horizontal wells

Manufacturer / Supplier: Brighter Star Drilling Fluids

Website: www.pcenigeria.com

Section 2: Hazards Identification

Hazard classification: Not classified as a hazardous chemical according to GB standards

- Dust formation may create a potential dust explosion hazard
- May cause mild irritation to eyes, skin, and respiratory tract

Section 3: Composition / Information on Ingredients

Chemical characterization: Positively charged mixed-layered metal hydroxide (MMH) crystal colloid, supplied as a dry powder.

Section 4: First-Aid Measures

Exposure Route	Action
Eye contact	Flush immediately with plenty of clean water for at least 15 minutes; seek medical attention if irritation persists
Skin contact	Wash the affected area with mild soap and water; launder contaminated clothing before reuse
Inhalation	Move to fresh air; administer oxygen if breathing is difficult; seek medical attention
Ingestion	Rinse mouth and drink water; do NOT induce vomiting; seek immediate medical attention

Section 5: Fire-Fighting Measures

Fire / explosion hazard: Fine dust dispersed in air may form an explosive dust–air mixture in the presence of an ignition source

Suitable extinguishing media: Water spray, dry chemical, or carbon dioxide (CO₂)

Hazardous combustion products: Thermal decomposition may yield carbon monoxide (CO), carbon dioxide (CO₂), and other hazardous gases

Precautions: Use non-sparking tools and prevent electrostatic discharge during fire response

Section 6: Accidental Release Measures

- Ensure adequate ventilation in the spill area
- Mechanically pick up the spilled material; avoid dry sweeping to minimize dust generation
- Prevent entry into sewers, surface water, or groundwater
- Dispose of contaminated material in accordance with local environmental regulations

Section 7: Handling and Storage

Handling: Avoid dust formation; use non-sparking tools and prevent electrostatic discharge. Provide adequate local exhaust ventilation. Wear appropriate personal protective equipment (see Section 8)

Storage: Store in a cool, dry, clean, and well-ventilated place at 5–40 °C (do not exceed 40 °C) and relative humidity ≤ 80%. Keep containers tightly closed to prevent moisture ingress; protect from direct sunlight, heat sources, and moisture. Do NOT store together with toxic, harmful, or polluting substances, or with strong oxidants

Shelf life: 12 months from the date of manufacture under proper sealed-storage conditions

Section 8: Exposure Controls / Personal Protection

Protection	Recommended Measures
Respiratory	MSHA-approved respirator or dust mask when handling the powder
Hands	Impervious rubber gloves
Eyes	Tightly fitting safety goggles with side shields (EN 166 / NIOSH compliant)
Body	Long-sleeve clothing; a Tyvek full-body suit is recommended for heavy exposure
Ventilation	Adequate local exhaust ventilation to prevent dust formation

Section 9: Physical and Chemical Properties

Property	Value
Appearance	White or light brown free-flowing powder
Loss on drying	≤ 15.0%

Property	Value
Screen residue (40-mesh)	$\leq 7.0\%$
Potential	≥ 35.0 mV
Acid solubility	$\geq 85.0\%$
Yield stress improvement rate	$\geq 300.0\%$

Section 10: Stability and Reactivity

Chemical stability: Stable under recommended storage conditions

Conditions to avoid: Dust formation, electrostatic discharge, ignition sources, heat, and moisture

Incompatible materials: Strong oxidants; toxic, harmful, or polluting substances (storage segregation)

Hazardous decomposition products: Carbon monoxide (CO), carbon dioxide (CO₂), and other hazardous gases from thermal decomposition

Section 11: Toxicological Information

Acute toxicity: Not classified as a hazardous chemical according to GB standards

Irritation: May cause mild irritation to eyes, skin, and respiratory tract

Section 12: Ecological Information

Environmental precautions: Prevent the product from entering sewers, surface water, or groundwater

Section 13: Disposal Considerations

- Dispose of the product and contaminated material in accordance with local, regional, and national environmental regulations
- Avoid dust generation during disposal operations

Section 14: Transport Information

Not regulated as dangerous goods for land, sea, or air transport. Protect from rain, moisture, and package damage during transit; use moisture-proof container liners for overseas shipment.

Section 15: Regulatory Information

Classification: Not classified as a hazardous chemical according to GB standards

Executive standard: Q/PSL038-2003

Section 16: Other Information

Disclaimer: All data, recommendations, and information contained in this document are provided for reference only. They are derived from laboratory testing and field experience obtained under specific conditions, and actual product performance may vary with formation characteristics, water quality, equipment, and operating practices. Users are advised to verify the suitability of this product through on-site trials and to adjust the dosage according to actual working conditions. In all cases, the actual conditions and results of use shall prevail. To the fullest extent permitted by applicable law, Brighter Star Drilling Fluids makes no warranties, express or implied, regarding the information herein, and assumes no responsibility or liability for any results, losses, or damages arising from the use of, or reliance on, this document.